Econ 101A
Section 10: Solutions

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Two things today. First we close the loop on present bias from last week's lecture, since Lecture 8 opens with polls on it. Then we meet the firm. The firm looks new, but most of it is the consumer problem with new labels. The production function plays the role of the utility function, isoquants play the role of indifference curves, and cost minimization is expenditure minimization. If you know Sections 3 to 7, you already know most of today.

1 Present Bias: When Today's Plan Is Not Tomorrow's Choice

The self at date t ranks streams of consumption with

$$u(c_t) + \frac{\beta}{1+\delta}\, u(c_{t+1}) + \frac{\beta}{(1+\delta)^2}\, u(c_{t+2}) + \ldots, \qquad 0 < \beta \leq 1.$$

With $\beta = 1$ this is standard exponential discounting. With $\beta < 1$, everything that is not *now* gets an extra discount β. The key point is that this extra discount falls on the next period only when the next period is in the future. Once it becomes today, it escapes the β.

Two dates, two selves. Take an activity with utility b at $t = 1$ and B at $t = 2$.

- Self 1 (it is now date 1): $b + \dfrac{\beta}{1+\delta} B$.
- Self 0 (looking ahead from date 0): $\dfrac{\beta}{1+\delta} b + \dfrac{\beta}{(1+\delta)^2} B = \dfrac{\beta}{1+\delta}\Big[b + \dfrac{1}{1+\delta} B \Big]$.

Self 0 weighs B against b with $1/(1+\delta)$. Self 1 weighs it with $\beta/(1+\delta)$, which is smaller. So the two selves can rank the same activity differently. This is **time inconsistency**: the plan made at date 0 is not the choice made at date 1.

The ice cream poll. $\delta = 0$, $\beta = 1/2$, $b = +10$ (taste today), $B = -16$ (stomach ache tomorrow).

$$\text{Self 0: } \tfrac{1}{2}(10) + \tfrac{1}{2}(-16) = 5 - 8 = -3, \qquad \text{Self 1: } 10 + \tfrac{1}{2}(-16) = 10 - 8 = +2.$$

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On Sunday you plan to skip the ice cream. On Monday you eat it.

Why we care. If you know your future self will deviate, you may want to tie its hands. That is a *demand for commitment*, which never shows up when $\beta = 1$. DellaVigna and Malmendier ("Paying Not to Go to the Gym," *AER* 2006) find that members on flat monthly contracts pay about \$17 per visit, when a ten-visit pass would have cost about \$10 per visit. People overestimate how often their future selves will go. In development, Duflo, Kremer and Robinson (*AER* 2011) show that Kenyan farmers who plan to buy fertilizer often fail to do so later, and that a small, time-limited offer of free delivery made right at harvest, when farmers have cash, raises fertilizer use substantially. Same model, real money.

2 Production Functions

A firm turns inputs $z = (z_1, \ldots, z_n)$ into output $y = f(z)$, with $f : \mathbb{R}^n_+ \to \mathbb{R}_+$. With two inputs we write $y = f(L, K)$. Standard assumptions:

- No free lunch: $f(0) = 0$.
- Positive marginal products: $f'_L > 0$, $f'_K > 0$.
- Diminishing marginal products: $f''_{LL} < 0$, $f''_{KK} < 0$. Each extra worker adds less, *holding capital fixed*.

Isoquants and the MRTS. The isoquant for output y is the set of input bundles with $f(L, K) = y$. Think of it as $K(L)$ and differentiate $f(L, K(L)) = y$ with respect to L:

$$f'_L + f'_K \, K'(L) = 0 \quad \Longrightarrow \quad \text{MRTS} \equiv -\frac{dK}{dL}\Big|_{f=y} = \frac{f'_L}{f'_K}.$$

This is the implicit function theorem again, exactly as for the MRS. The MRTS says how much capital you can give up for one more worker and keep output constant.

One difference with utility. Utility is ordinal: u and u^2 describe the same person. A production function is cardinal: output is counted in cars or chips, so f and f^2 are two different technologies. That is why returns to scale, below, is a meaningful idea for firms and not for consumers.

3 Returns to Scale

Scale *all* inputs by the same factor $t > 1$. For every z:

$$\text{DRS: } f(tz) < tf(z), \qquad \text{CRS: } f(tz) = tf(z), \qquad \text{IRS: } f(tz) > tf(z).$$

For Cobb–Douglas $f(L, K) = AK^\alpha L^\beta$:

$$f(tL, tK) = A(tK)^\alpha (tL)^\beta = t^{\alpha+\beta} AK^\alpha L^\beta = t^{\alpha+\beta} f(L, K),$$

so returns to scale are decreasing, constant or increasing as $\alpha + \beta$ is below, equal to, or above 1.

Do not mix this up with diminishing marginal products. Diminishing f'_L is about adding labor *alone*. Returns to scale are about adding *everything* together. Exercise 2 shows a technology with both diminishing marginal products and increasing returns.

4 Cost Minimization, Step 1

The firm wants maximal profit. We split the problem in two.

1. For a given output y, find the cheapest input bundle. (Today.)
2. Given the resulting cost, pick the best y. (Next section.)

With wage w and rental rate of capital r, step 1 is

$$\min_{L,K} \; wL + rK \quad \text{s.t.} \quad f(L,K) \geq y.$$

This is expenditure minimization with new names:

Consumer	Firm
utility $u(x_1, x_2)$	production $f(L,K)$
target utility $\bar{u}$	target output y
prices p_1, p_2	input prices w, r
MRS $= p_1/p_2$	MRTS $= w/r$
Hicksian demand $h(p, \bar{u})$	conditional factor demand $L^*(w,r,y)$
expenditure function $e(p, \bar{u})$	cost function $c(w,r,y)$
Shephard: $\partial e/\partial p_1 = h_1$	Shephard: $\partial c/\partial w = L^*$

The constraint binds if $w, r > 0$ and f is increasing. Producing more than y means you could cut an input and save money.

First-order conditions. With $\mathcal{L} = wL + rK - \lambda\left[f(L,K) - y\right]$:

$$w = \lambda f'_L, \qquad r = \lambda f'_K \quad \Longrightarrow \quad \frac{f'_L}{f'_K} = \frac{w}{r}.$$

At the optimum the isoquant is tangent to the lowest isocost line $wL + rK = C$. Solving gives the conditional factor demands $L^*(w,r,y)$, $K^*(w,r,y)$, and the value of the problem is the cost function

$$c(w,r,y) = wL^*(w,r,y) + rK^*(w,r,y).$$

The multiplier λ is $\partial c/\partial y$: marginal cost. Keep that in mind for next time.

5 Exercises

Exercise 1 (present bias and procrastination) Let $\delta = 0$ and $\beta = 1/2$. Doing the problem set at $t = 1$ costs 6 utils at $t = 1$ and brings a benefit of 10 utils at $t = 2$ (a better grade). Not doing it gives 0.

(a) From the point of view of self 0, should the problem set get done at $t = 1$?

(b) Does self 1 actually do it?

(c) For which values of β does self 1 do it? For which values do the two selves disagree?

(d) At $t = 0$ you can sign up for a study group that charges a penalty P at $t = 1$ if you skip the problem set. What is the smallest P that gets the problem set done? How much would self 0 pay at $t = 0$ to join?

Solution. **(a)** With $\delta = 0$, self 0 puts weight $\beta/(1+\delta) = \beta = \frac{1}{2}$ on $t = 1$ and $\beta/(1+\delta)^2 = \beta = \frac{1}{2}$ on $t = 2$. So

$$U_0(\text{do}) = \tfrac{1}{2}(-6) + \tfrac{1}{2}(10) = -3 + 5 = 2 > 0 = U_0(\text{skip}).$$

Self 0 wants the problem set done.

(b) Self 1 puts weight 1 on $t = 1$ and $\beta/(1+\delta) = \frac{1}{2}$ on $t = 2$:

$$U_1(\text{do}) = -6 + \tfrac{1}{2}(10) = -6 + 5 = -1 < 0 = U_1(\text{skip}).$$

Self 1 skips it. The plan from Sunday falls apart on Monday.

(c) Self 1 does it when

$$-6 + 10\beta \geq 0 \iff 10\beta \geq 6 \iff \beta \geq 0.6.$$

Self 0 wants it done when $\beta(-6) + \beta(10) = 4\beta > 0$, which holds for every $\beta > 0$. So the selves disagree exactly when $0 < \beta < 0.6$. With $\beta = 1$ there is never a disagreement.

(d) With the penalty, self 1 compares doing it, -1, with skipping and paying, $-P$. Self 1 does it when

$$-1 \geq -P \iff P \geq 1.$$

So the smallest penalty is $P = 1$. With $P \geq 1$ the problem set gets done and the penalty is never paid. Self 0 then gets $U_0 = 2$ instead of 0. Paying m at $t = 0$ costs self 0 exactly m (weight 1 on today), so self 0 would pay up to $m = 2$ utils to join. This is the value of commitment. Under $\beta = 1$ it would be zero, because self 1 would do the problem set anyway.

Exercise 2 (Cobb–Douglas technology) Let $y = f(L, K) = AK^{\alpha}L^{\beta}$ with $A, \alpha, \beta > 0$.

(a) Compute f'_L and f''_{LL}. When is the marginal product of labor diminishing?

(b) Compute the MRTS. Show that isoquants are convex.

(c) Find the returns to scale.

(d) Let $\alpha = \beta = 0.6$. Are marginal products diminishing? What are the returns to scale? Explain why there is no contradiction.

Solution. **(a)** Treat K as a constant:

$$f'_L = \beta A K^{\alpha} L^{\beta-1} > 0, \qquad f''_{LL} = \beta(\beta-1) A K^{\alpha} L^{\beta-2}.$$

Since $\beta > 0$, $A > 0$ and $K^{\alpha}L^{\beta-2} > 0$, the sign of f''_{LL} is the sign of $\beta - 1$. The marginal product of labor is diminishing if and only if $\beta < 1$. By the same steps, $f''_{KK} = \alpha(\alpha-1)AK^{\alpha-2}L^{\beta} < 0$ if and only if $\alpha < 1$.

(b) We need $f'_K = \alpha A K^{\alpha-1} L^{\beta}$. Then

$$\text{MRTS} = \frac{f'_L}{f'_K} = \frac{\beta A K^{\alpha} L^{\beta-1}}{\alpha A K^{\alpha-1} L^{\beta}} = \frac{\beta}{\alpha} K^{\alpha-(\alpha-1)} L^{(\beta-1)-\beta} = \frac{\beta}{\alpha} K^{1} L^{-1} = \frac{\beta}{\alpha}\frac{K}{L}.$$

For convexity, write the isoquant $AK^{\alpha}L^{\beta} = y$ as K in terms of L:

$$K^{\alpha} = \frac{y}{AL^{\beta}} \quad \Longrightarrow \quad K(L) = \Big(\frac{y}{A}\Big)^{1/\alpha} L^{-\beta/\alpha}.$$

Differentiate once:

$$K'(L) = -\frac{\beta}{\alpha}\Big(\frac{y}{A}\Big)^{1/\alpha} L^{-\beta/\alpha-1} < 0.$$

(Check: $K'(L) = -\frac{\beta}{\alpha}\big[(y/A)^{1/\alpha}L^{-\beta/\alpha}\big]\frac{1}{L} = -\frac{\beta}{\alpha}\frac{K}{L} = -\text{MRTS}$.) Differentiate again. The exponent $-\beta/\alpha - 1$ comes down and the minus signs cancel:

$$K''(L) = -\frac{\beta}{\alpha}\Big(\frac{y}{A}\Big)^{1/\alpha}\Big(-\frac{\beta}{\alpha}-1\Big)L^{-\beta/\alpha-2} = \frac{\beta}{\alpha}\Big(\frac{\beta}{\alpha}+1\Big)\Big(\frac{y}{A}\Big)^{1/\alpha} L^{-\beta/\alpha-2} > 0.$$

The isoquant is downward sloping and convex for any $\alpha, \beta > 0$.

(c) For $t > 1$:

$$f(tL, tK) = A(tK)^{\alpha}(tL)^{\beta} = A\, t^{\alpha} K^{\alpha}\, t^{\beta} L^{\beta} = t^{\alpha+\beta} f(L, K).$$

Compare $t^{\alpha+\beta}$ with t. For $t > 1$, $t^{\alpha+\beta} > t$ iff $\alpha + \beta > 1$. So IRS if $\alpha + \beta > 1$, CRS if $\alpha + \beta = 1$, DRS if $\alpha + \beta < 1$.

(d) Each exponent is $0.6 < 1$, so by (a) both marginal products are diminishing. But $\alpha + \beta = 1.2 > 1$, so by (c) there are increasing returns. Doubling both inputs multiplies output by $2^{1.2} \approx 2.30$. No contradiction: a diminishing marginal product of labor holds capital fixed, so each new worker has less capital to work with. Under returns to scale, capital grows with labor, so that effect is gone.

Exercise 3 (classify the returns to scale) For each technology, compute $f(tL, tK)$ for $t > 1$ and classify.

(a) $f = 2L + 3K$

(b) $f = \min\{L, 2K\}$

(c) $f = \sqrt{L} + \sqrt{K}$

(d) $f = (LK)^{2/3}$

(e) $f = L + K^2$ (careful)

Solution. **(a)** $f(tL, tK) = 2tL + 3tK = t(2L + 3K) = tf$. CRS.

(b) $f(tL, tK) = \min\{tL, 2tK\} = t\min\{L, 2K\} = tf$, since multiplying both entries by $t > 0$ does not change which is smaller. CRS.

(c) $f(tL, tK) = \sqrt{tL} + \sqrt{tK} = \sqrt{t}\sqrt{L} + \sqrt{t}\sqrt{K} = t^{1/2} f$. For $t > 1$, $t^{1/2} < t$, so $f(tz) < tf(z)$ whenever $f(z) > 0$. DRS.

(d) $f(tL, tK) = (tL \cdot tK)^{2/3} = (t^2 LK)^{2/3} = t^{4/3}(LK)^{2/3} = t^{4/3} f$. Since $4/3 > 1$, IRS.

(e) $f(tL, tK) = tL + t^2K^2$ while $tf = tL + tK^2$. Subtract:

$$f(tz) - tf(z) = t^2K^2 - tK^2 = t(t-1)K^2.$$

For $t > 1$ this is strictly positive when $K > 0$ and exactly zero when $K = 0$. So returns are increasing at every bundle that uses capital and constant along the axis $K = 0$. The definition asks for "all z," so this technology fits none of the three labels. Returns to scale can change from one bundle to another.

Exercise 4 (cost minimization, step 1) Let $f(L, K) = K^{1/2}L^{1/2}$, with $w, r > 0$ and target output $y > 0$.

(a) Explain why the constraint binds. Then give a counterexample when $w = r = 0$, and one with a lumpy technology $f(L, K) = 10$ if $L \geq 1$, $f = 0$ otherwise, for $y = 4$.

(b) Write the Lagrangian and the first-order conditions. Show that $K = (w/r)L$ at the optimum.

(c) Solve for the conditional factor demands $L^*(w, r, y)$ and $K^*(w, r, y)$.

(d) Find the cost function $c(w, r, y)$.

(e) Check Shephard's lemma, $\partial c/\partial w = L^*$, and check that $\lambda = \partial c/\partial y$. Sign $\partial L^*/\partial w$ and $\partial L^*/\partial r$.

Solution. **(a)** Suppose the optimum (L^*, K^*) has $f(L^*, K^*) > y$. Since $y > 0 = f(0, 0)$, at least one input is positive, say $L^* > 0$. Because f is continuous, cutting L a little keeps output at least y. The cost falls by w times the cut, which is strictly positive because $w > 0$. So (L^*, K^*) was not the cheapest bundle, a contradiction. Hence $f(L^*, K^*) = y$.

Counterexample 1: if $w = r = 0$, every bundle costs zero. The bundle $L = K = 100$ gives output $100 > y$ and is optimal (for $y < 100$). The constraint need not bind.

Counterexample 2: with the lumpy technology, any $L < 1$ produces 0, which is not enough. The cheapest feasible bundle is $L = 1$, $K = 0$, with cost w. It produces $10 > 4$. The constraint is slack because the technology jumps.

(b) The Lagrangian is

$$\mathcal{L} = wL + rK - \lambda\big(K^{1/2}L^{1/2} - y\big).$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial L} = w - \lambda\, \tfrac{1}{2} K^{1/2}L^{-1/2} = 0, \qquad \frac{\partial \mathcal{L}}{\partial K} = r - \lambda\, \tfrac{1}{2} K^{-1/2}L^{1/2} = 0, \qquad K^{1/2}L^{1/2} = y.$$

Move the λ terms to the right and divide the first condition by the second. The λ and the $\frac{1}{2}$ cancel:

$$\frac{w}{r} = \frac{K^{1/2}L^{-1/2}}{K^{-1/2}L^{1/2}} = K^{1/2+1/2}L^{-1/2-1/2} = \frac{K}{L} \quad \Longrightarrow \quad K = \frac{w}{r}L.$$

(c) Put $K = (w/r)L$ into the constraint:

$$\Big(\frac{w}{r}L\Big)^{1/2} L^{1/2} = y \implies \Big(\frac{w}{r}\Big)^{1/2} L^{1/2}L^{1/2} = y \implies \Big(\frac{w}{r}\Big)^{1/2} L = y \implies L^* = y\Big(\frac{r}{w}\Big)^{1/2}.$$

Then

$$K^* = \frac{w}{r}L^* = \frac{w}{r}\,y\Big(\frac{r}{w}\Big)^{1/2} = y\Big(\frac{w}{r}\Big)^{1}\Big(\frac{w}{r}\Big)^{-1/2} = y\Big(\frac{w}{r}\Big)^{1/2}.$$

(d)

$$c(w,r,y) = wL^* + rK^* = w\,y\,r^{1/2}w^{-1/2} + r\,y\,w^{1/2}r^{-1/2} = y\,w^{1/2}r^{1/2} + y\,w^{1/2}r^{1/2} = 2y\sqrt{wr}.$$

(e) Shephard's lemma. Differentiate $c = 2y\,w^{1/2}r^{1/2}$ in w:

$$\frac{\partial c}{\partial w} = 2y\cdot\tfrac{1}{2}w^{-1/2}r^{1/2} = y\Big(\frac{r}{w}\Big)^{1/2} = L^*.$$

The multiplier. From the first FOC, $\lambda = \dfrac{w}{\frac{1}{2}K^{1/2}L^{-1/2}} = 2w\Big(\dfrac{L}{K}\Big)^{1/2}$. At the optimum $\dfrac{L^*}{K^*} = \dfrac{y(r/w)^{1/2}}{y(w/r)^{1/2}} = \dfrac{r}{w}$, so

$$\lambda = 2w\Big(\frac{r}{w}\Big)^{1/2} = 2w^{1/2}r^{1/2} = 2\sqrt{wr} = \frac{\partial c}{\partial y}.$$

The shadow price of output is marginal cost. Comparative statics from $L^* = y\,r^{1/2}w^{-1/2}$:

$$\frac{\partial L^*}{\partial w} = -\tfrac{1}{2}\,y\,r^{1/2}w^{-3/2} < 0, \qquad \frac{\partial L^*}{\partial r} = \tfrac{1}{2}\,y\,r^{-1/2}w^{-1/2} > 0.$$

Holding output fixed, the firm moves away from the input that got more expensive. Like Hicksian demand, there is no income effect here: only substitution.